

NALLABATI VENKATA DURGA SAI SARAT CHANDRA

Target Role: Research Scientist / Engineer - Scientific Machine Learning • Google DeepMind / NVIDIA Modulus

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EDUCATION & NATIONAL MERIT PEDIGREE

Amrita Vishwa Vidyapeetham

August 2024 – Expected May 2028

B.Tech in Robotics & Artificial Intelligence (Cumulative GPA: 9.05 / 10.00, Top 5% Cohort)

Amritapuri, India

- Academic Standing: AP POLYCET State Merit Rank: 497 (Top 0.5% Statewide Percentile).
- National Distinctions: ISRO-IIRS YUVIKA (Selected for space science analytics) | NAEST National Physics Test (Cleared Prelims & Finals, IAPT / Prof. H.C. Verma) | MTSO Olympiad (Level 2 Qualifier).

PEER-REVIEWED PUBLICATIONS & PROCEEDINGS

- N. V. D. S. Sarat Chandra, M. Nunna, Asha C. A., P. C. Sreekumar, “Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity.” in Proc. IEEE ICRM, 2025.
- N. V. D. S. Sarat Chandra, et al., “Physics-Guided Diffusion with Thermodynamic Awareness for 3D Molecular Generation.” Submitted to IEEE IES, 2026.
- N. V. D. S. Sarat Chandra, et al., “Domain-Aware Multi-Modal Industrial Diagnostics: Integrating Mechanical Heuristics with Conformal Uncertainty.” Submitted 2026.

TARGETED ENGINEERING PROJECTS & SYSTEMS ARCHITECTURE

Dynamic Diffusion: Physics-Guided 3D Molecular Generation

Top Finalist, IEEE IES Generative AI Challenge

Score-Based Diffusion, PyTorch, Thermodynamic Priors, SE(3) Equivariance

- Core Impact: Embedded Gibbs free energy surfaces & Hamiltonian conservation priors in reverse-time SDEs.

Blood Neural Network: PINNs for Inverse Hemodynamic Rheology Discovery

SciML Researcher

PINNs, Navier-Stokes, Carreau-Yasuda, ANSYS Fluent CFD, NTK Trace Balancing

- Core Impact: <2.4% relative L2 error vs high-resolution ANSYS Fluent pulsatile CFD benchmarks.

Domain-Aware Industrial Failure Prediction & Diagnostics Framework

PIML Researcher

Mechanical Vibration Heuristics, XGBoost/Ensemble, Optuna, Conformal Uncertainty

- Core Impact: 85.2% recall at 91.4% precision (0.94 AUROC); modeled 68% downtime reduction.

TECHNICAL SKILLS & CORE COMPETENCIES

Languages & Runtimes: C (C99/C11), C++ (C++17/20), Python (3.12), Bash, MATLAB, POSIX Sockets, FreeRTOS, Bare-Metal

Robotics, AI & Middleware: ROS 2 (Jazzy), CycloneDDS, Forward/Inverse Kinematics, OpenCV, PyTorch, PINNs, ONNX Runtime, Vitis AI EP

Hardware, Buses & Silicon: CAN Bus (2.0B), UART DMA, I2C (9-clock recovery), SPI, Valettron A7670C (4G LTE), STM32, ESP32, Raspberry Pi 5

Simulation & Tooling: ANSYS Fluent (CFD), SolidWorks, Fusion 360, Linux (Ubuntu Server), Docker, Git/GitHub, LaTeX